

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop high quality solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#)
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#)
Thesis: *Evaluating land degradation using image processing methods*

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Geospatial QA & Analysis:** Developed a custom [Python program](#) using **pandas** and **rasterio** to process drone-captured location datasets. Integrated camera and sensor calibration parameters to ensure data accuracy, utilizing Moran's I coefficient for spatial anomaly detection and clustering.
- **Data Visualization:** Conducted exploratory data analysis (EDA) and visualized complex spatial datasets in **QGIS** to validate model outputs and identify inconsistencies in land degradation metrics.
- **Reproducibility & Integrity:** Engineered a unified, documented data architecture for research backups, facilitating automated data retrieval and version control. This system ensured the consistency and completeness of large-scale datasets for multi-year research.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Self-initiated project to reproduce key examples from C. Bishop and H. Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of **polynomial regression** and **probability theory**.

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a custom JSON parser to extract and sensor data obtained by aiSim, an autonomous driving simulator, into a nerfstudio format.

SKILLS

Programming: Python, Julia, C/C++, R, MATLAB
Tools: Docker, Linux, Bash, QGIS, Git, $\text{\LaTeX}$